

Root-of-Unity Packets that Isolate a Single Power Moment

Bin Wang

July 2026

Abstract

We construct a genuine finite normal spectrum that changes one prescribed power moment without changing any lower moment. The packet consists of a complete root set of a real monomial equation, repeated with integer multiplicity. Its entire moment sequence, radius, rank, and squared mass are explicit. This supplies the triangular atom needed for exact finite-prefix spectral realization, while making no identification with the actual noisy operator.

1 Packet definition

Fix $d, L \geq 1$ and $w \in \mathbb{R} \setminus \{0\}$. Let

$$\mathcal{P}_d(w, L) = \{\mu : \mu^d = w/(dL)\},$$

with every root repeated L times. Since the defining polynomial has real coefficients, the multiset is closed under complex conjugation.

Theorem 1.1 (moment isolation). *The packet has rank dL , radius*

$$r_d = (|w|/(dL))^{1/d},$$

and power sums

$$p_n(\mathcal{P}_d) = 0 \quad (d \nmid n), \quad p_{md}(\mathcal{P}_d) = \frac{w^m}{(dL)^{m-1}}.$$

In particular, $p_n = 0$ for $n < d$ and $p_d = w$. Its squared spectral mass is

$$M_2 = dLr_d^2.$$

Proof. Write the roots as $r_d e^{i(\theta+2\pi j)/d}$. The roots-of-unity sum vanishes unless d divides n . At $n = md$, summing and multiplying by L gives $Ld(w/(dL))^m$, which is the stated formula. The geometric quantities are immediate. $\square$

Corollary 1.2 (modulus cap). *For $q > 0$, if*

$$L \geq \frac{|w|}{dq^d},$$

then every packet root satisfies $|\mu| \leq q$.

Remark 1.3. The packet is an unweighted finite spectrum with integer multiplicities, not merely a positive moment measure. It is nevertheless synthetic: no theorem identifies it with eigenvalues of the noisy transfer operator. Gates A–E remain false/open.